

The Complex Body

Complex Analysis

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Definition

A group $\mathcal{G}$ is a set of elements $\{X, Y, \dots\}$, together with a rule for combining them that associates with each ordered pair X, Y a 'product' or combination law $X \bullet Y$ for which the following conditions must be satisfied.

- i) For every pair of elements X, Y that belongs to $\mathcal{G}$, the product $X \bullet Y$ also belongs to $\mathcal{G}$. (This is known as the **closure property**)
- ii) For all triples X, Y, Z the **associative law** holds; in symbols,

$$X \bullet (Y \bullet Z) = (X \bullet Y) \bullet Z.$$

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iii) *There exists a unique element I , belonging to $\mathcal{G}$, with the property that*

$$I \bullet X = X = X \bullet I$$

*for all X belonging to $\mathcal{G}$. This element is known as the **identity element** of the group.*

iv) *For every element X of $\mathcal{G}$, there exists an element X^{-1} , also belonging to $\mathcal{G}$, such that*

$$X^{-1} \bullet X = I = X \bullet X^{-1}.$$

*X^{-1} is called the **inverse** of X .*

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Theorem 1.1

The identity of any particular group and the inverse of any particular element is unique.

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Theorem 1.2

The inverse of a product of elements in a group is the product of the inverses in reverse order.

Theorem 1.3

Given any pair of elements X, Y belonging to $\mathcal{G}$, there exist unique elements U, V , also belonging to $\mathcal{G}$, such that

$$X \bullet U = Y \quad \text{and} \quad V \bullet X = Y.$$

Theorem 1.4 (Cancellation Law)

The **cancellation law** can be stated as follows. If

$$X \bullet Y = X \bullet Z$$

for some X belonging to $\mathcal{G}$, then $Y = Z$. Similarly,

$$Y \bullet X = Z \bullet X$$

implies the same conclusion.

Definition

If any two particular elements of a group satisfy

$$X \bullet Y = Y \bullet X,$$

*they are said to **commute** under the operation $\bullet$; if all pairs of elements in a group satisfy such condition, then the group is said to be **Abelian**. Abelian groups are usually denoted by $(\mathcal{G}, +)$ to emphasise the commutativity of the operation defining the group.*

Definition

Suppose $\mathcal{R}$ is a set in which a relation of equality, denoted by $=$, and operations of addition and multiplication, denoted by $+$ and $\cdot$, respectively, are defined. Then $\mathcal{R}$ is a **ring** (with respect to these operations) if these conditions hold:

- i) $\mathcal{R}$ forms an Abelian group with respect to addition.
- ii) $\mathcal{R}$ is closed with respect to an associative multiplication.
- iii) Two distributive laws hold in $\mathcal{R}$: $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$ for all x, y, z in $\mathcal{R}$.

Definition

Let $\mathcal{R}$ be a ring. Then the identity of $(\mathcal{R}, +)$, denoted by 0 , is called the **additive identity** and referred to as the **zero** of $\mathcal{R}$ and has the defining property $x + 0 = x$ for all $x \in \mathcal{R}$. If $y \in \mathcal{R}$, then the inverse (in the sense of the Abelian group) $-y$ is called the **additive inverse** of y or the **negative** of y and has the defining property $y + (-y) = 0$. Furthermore, if $x, y \in \mathcal{R}$, then the **subtraction** in $\mathcal{R}$ is defined by

$$x - y := x + (-y).$$

Definition

Let $\mathcal{R}$ be a ring. If there exists an element e in $\mathcal{R}$ such that $x \cdot e = e \cdot x = x$ for all x in $\mathcal{R}$, then e is called a **unity**, and $\mathcal{R}$ is said to be a **ring with unity**. If multiplication in $\mathcal{R}$ is commutative, then $\mathcal{R}$ is called a **commutative ring**.

Definition

Let $\mathcal{R}$ be a ring with unity e , and let $a \in \mathcal{R}$. If there is an element x in $\mathcal{R}$ such that $ax = xa = e$, then x is a **multiplicative inverse** of a and a is called a **unit** (or an **invertible element**) in $\mathcal{R}$.

Definition

Let $\mathbb{K}$ be a ring. Then $\mathbb{K}$ is said to be a **body** (or a **field**) if the following conditions hold.

- i) $\mathbb{K}$ is a commutative ring.
- ii) $\mathbb{K}$ has a unity e , and $e \neq 0$.
- iii) Every nonzero element of $\mathbb{K}$ has a multiplicative inverse.

Example 1.5

The set $\mathbb{R}$ of real numbers forms a body under the usual definitions of equality, sum, and multiplication.

Definition

The set $\mathbb{C}$ of **complex numbers** is the set of ordered pairs (x, y) of real numbers x and y , equipped with algebraic operations of **addition** and **multiplication**:

$$(x, y) + (x', y') = (x + x', y + y'),$$

$$(x, y)(x', y') = (xx' - yy', xy' + x'y).$$

A **complex number** is an element of $\mathbb{C}$. For a complex number $z = (x, y)$, x is called the **real part** of z , denoted by $\Re(z)$ and y is called the **imaginary part**, denoted by $\Im(z)$.

Definition

*Two complex numbers z_1 and z_2 are said to be **equal** if and only if $\Re(z_1) = \Re(z_2)$ and $\Im(z_1) = \Im(z_2)$.*

Definition

A complex number z is said to be **purely real** [respectively, **purely imaginary**], if $\Im(z) = 0$ [respectively, $\Re(z) = 0$]. The **imaginary unit** is the complex number $(0, 1)$ and is denoted by i .

Definition

Given that

$$(x, y) = (x, 0) + (0, y) = x(1, 0) + y(0, 1),$$

*we usually write $(x, y) \equiv x + iy$, and term this the **algebraic form** of a complex number. In this sense, the number 1 is the complex number $(1, 0)$.*

Definition

*If z is a complex number, the number obtained by changing the sign of its imaginary part is called the **complex conjugate**, or simply **conjugate**, of z and is denoted by the symbol z^* (or $\bar{z}$). In other words, if $z = a + ib$, then its conjugate is $z^* = a - ib$.*

Definition

If z is a nonzero complex number, we define:

$$\frac{1}{z} = \frac{z^*}{|z|^2}.$$

Definition

The **zero** in the complex numbers is the number $0 + 0i$ and the **unity** is $1 + 0i$. The zero and unity are denoted by 0 and 1 , respectively. The zero is the additive identity in the complex numbers since, $z + 0 = z$. Similarly, the unity is the multiplicative identity since, $z \cdot 1 = z$.

Definition

*If z is a complex number, then the **modulus** of z is denoted by $|z|$ and defined as the positive square root of the (real) number z^*z .*

Proposition 1.6

The set $\mathbb{C}$ of complex numbers equipped with the previously defined relation of equality ($=$) and operations $(+, \cdot)$ forms a body (with respect to these operations).

*We will refer to $\mathbb{C}$ as the **complex body**.*